Software Requirements Specification

for

Laundry Management System

Version 1.0 approved

Documented By :

24MIS7056 PAPPURU NIKHILA

24MIS7060 KARLAPUDI SAI THANUJA

24MIS7064 KARUMURI CHARAN DEEP REDDY

24MIS7069 THANGELLAPALLI NIKHITHA

24MIS7070 JONNALAGADDA SNEHITHA

VIT-AP UNIVERSITY

Date Created : 08 August,2025

Copyright © 1999 by Karl E. Wiegers. Permission is granted to use, modify, and distribute this document.

Table of Contents

Table of Contents ii
Revision History ii
1. Introduction 1
1.1 Purpose 1
1.2 Document Conventions 1
1.3 Intended Audience and Reading Suggestions 1
1.4 Product Scope 1
1.5 References 2
2. Overall Description 2
2.1 Product Perspective 2
2.2 Product Functions 2
2.3 User Classes and Characteristics 2
2.4 Operating Environment 2
2.5 Design and Implementation Constraints 3
2.6 User Documentation 3
2.7 Assumptions and Dependencies 3
3. External Interface Requirements 3
3.1 User Interfaces 3
3.2 Hardware Interfaces 3
3.3 Software Interfaces 3
3.4 Communications Interfaces 4
4. System Features 4
4.1 User Registration and Login 4
4.2 Place Order Feature 5
4.3 Submit Complaint 6
4.4 Return Laundry 7
4.5 Manage Queries 8
5. Other Nonfunctional Requirements 8
5.1 Performance Requirements 8
5.2 Safety Requirements 9
5.3 Security Requirements 9
5.4 Software Quality Attributes 9
5.5 Business Rules 9
6. Other Requirements 9
Appendix A: Glossary 9
Appendix B: Analysis Models 10
Appendix C: To Be Determined List 10

Revision History

Name	Date	Reason For Changes	Version
First Edition	08-08-25	Initial draft	1.0
Final Edition	25-08-25	Final draft	1.1

1. Introduction

1.1 Purpose

The Laundry Management System is a web-based application designed for hostel students and laundry managers to manage the process of placing laundry orders, assigning tasks, and returning clothes on schedule.

This SRS specifies all functional, non-functional, and interface requirements for the system. The software will be used by students (users) to place orders and complaints, and by managers/admins to manage laundry operations.

1.2 Document Conventions

Bold text indicates major system modules.
All requirement IDs are labeled as REQ-x.y for tracking.
Dates are written in DD-MM-YYYY format.
Priority levels: H = High, M = Medium, L = Low.

1.3 Intended Audience and Reading Suggestions

Developers: For implementing the described functionalities.

Project Managers: To understand scope and deliverables.

Testers: To prepare test cases based on system features.

End Users: To understand what features will be available. Reading order suggestion: Start with Section 2 (Overall Description) for a project overview, then read Section 4 (System Features) for specific functional requirements.

1.4 Product Scope

The Laundry Management System enables:

Students to place laundry orders with service type (gentle/daily/hard wash), select detergent, and specify room number.

Managers to assign staff to collect, wash, and return clothes on fixed schedules.

Admins to manage complaints and maintain detergent stock. Benefits:

Eliminates manual record-keeping.

Ensures timely laundry service.

Improves communication between users and managers.

1.5 References

IEEE SRS Standard (IEEE 830-1998).

College Hostel Policy Document (Laundry Services Section).

[Sample Hostel Laundry Web Portal UI Guidelines].

2. Overall Description

2.1 Product Perspective

The system is new and self-contained, replacing manual registers with a database-driven web application. It interacts with:

Users via a web UI.

A MySQL database for order storage.

2.2 Product Functions

User login using hostel credentials.

Apply order (service type, detergent selection, clothes count).

Apply complaint (with photo and description).

Admin view & update orders.

Fixed pickup and return dates.

Block-wise manager contacts.

2.3 User Classes and Characteristics

Student User: Basic computer skills, occasional use.

Admin Manager: Familiar with order management, daily use.

2.4 Operating Environment

Platform: Web application.

Server: Apache/Nginx.

Database: MySQL.

Client: Chrome, Firefox, Edge browsers.

OS: Windows/Linux for server.

2.5 Design and Implementation Constraints

Must integrate with hostel's login system (e.g., VTOP).

Database hosted on campus intranet (no public access).

Pickup/Return dates fixed by system logic.

2.6 User Documentation

User manual with screenshots.

Quick start guide for managers.

Online help section.

2.7 Assumptions and Dependencies

Laundry staff availability is constant.

Hostel internet connectivity is stable.

Students will have access to PCs or mobile devices.

3. External Interface Requirements

3.1 User Interfaces

Login Page: Username, password, block selector.

User Dashboard: Apply Order, Apply Complaint, View Orders.

Admin Dashboard: View Orders, View Complaints.

3.2 Hardware Interfaces

Server hardware (minimum 8 GB RAM, 4-core CPU).

3.3 Software Interfaces

HTML5, CSS3, JavaScript (optional: Bootstrap for styling).

MySQL database connection.

3.4 Communications Interfaces

HTTP/HTTPS protocol for client-server communication.

Internal LAN or campus Wi-Fi.

4. System Features

4.1 User Registration and Login

4.1.1 Description and Priority

The system shall provide a secure registration and login mechanism for customers, managers, and admins.

***Purpose**: To allow authorized users to access system features according to their roles.*

***Priority**: High — Without this feature, no other system functionality can be accessed.*

4.1.2 Stimulus/Response Sequences

Stimulus:

1. *User selects "Register" from the home page.*
2. *User enters required details (name, email, phone number, password).*
3. *User clicks "Submit" for registration.*
4. *User selects "Login" and enters username and password.*
5. *User clicks "Login".*

Response:

1. *The system validates inputs and stores registration details in the database.*
2. *On login, the system authenticates credentials against stored records.*
3. *If credentials are correct, the system redirects the user to the appropriate dashboard based on their role.*
4. *If credentials are invalid, an error message is displayed.*

4.1.3 Functional Requirements

***FR1**: Display registration form with fields for name, email, phone number, and password, and store details securely in the database.*

***FR2**: Display login form for username and password, and authenticate credentials.*

***FR3**: Implement role-based access control (Customer, Manager, Admin) and redirect each role to its respective dashboard.*

***FR4**: Encrypt passwords before storing in the database.*

4.2 Place Order Feature

4.2.1 Description and Priority

The system shall allow customers to place laundry orders by selecting wash type, detergent preference, collection date, and return date.

***Purpose**: To collect all necessary details for processing laundry orders.*

***Priority**: High — It is the core functionality of the system.*

4.2.2 Stimulus/Response Sequences

***Stimulus**:*

1. User logs in and selects "Place Order".

2. User selects wash type (Gentle Wash, Daily Wash, Hard Wash).

3. User selects detergent type (optional feature).

4. User enters the number and type of clothes.

5. User submits the order.

***Response**:*

1. System validates inputs and stores order details in the database.

2. System automatically calculates collection and return dates based on the order date.

3. System generates an order ID and displays confirmation to the user.

4.2.3 Functional Requirements

***FR1**: Provide a form for order placement with wash type, detergent choice, and clothes count/type.*

***FR2**: Calculate collection date as the next day and return date as the day after collection.*

***FR3**: Store order details linked to the user's account in the database.*

***FR4**: Display confirmation page with order ID, dates, and order summary.*

4.3 Submit Complaint

4.3.1 Description and Priority

The system shall allow users to submit complaints regarding lost clothes, delays, or service issues.

***Purpose**: To provide a communication channel between customers and management.*

***Priority**: Medium — Enhances customer satisfaction but not core to order processing.*

4.3.2 Stimulus/Response Sequences

***Stimulus**:*

1. User logs in and selects "Submit Complaint".

2. User fills complaint form with issue description and order ID (if applicable).

3. User submits the complaint.

***Response**:*

1. System stores the complaint in the database with a unique complaint ID.

2. System notifies the manager for review.

3. System confirms submission to the user.

4.3.3 Functional Requirements

***FR1**: Provide complaint submission form with order ID and description fields.*

***FR2**: Store complaints in the database with timestamps and user details.*

***FR3**: Notify managers automatically upon complaint submission.*

***FR4**: Display complaint status to the user.*

4.4 Return Laundry

4.4.1 Description and Priority

The system shall manage the process of returning cleaned clothes to customers on the scheduled return date.

***Purpose**: To ensure timely delivery of cleaned laundry.*

***Priority**: High — Affects customer satisfaction and service reliability.*

4.4.2 Stimulus/Response Sequences

***Stimulus**:*

1. Manager selects "View Orders for Return" in the dashboard

2. Manager marks orders as "Ready for Return" once clothes are ready.

3. Clothes are delivered to the customer.

***Response**:*

1. System updates the order status to "Returned".

2. System logs return date and time in the database.

3. System notifies customer that laundry has been returned.

4.4.3 Functional Requirements

***FR1**: Provide list of orders due for return on the current date.*

***FR2**: Allow manager to mark orders as returned.*

***FR3**: Update order records in the database with return status and date.*

***FR4**: Send notification to the customer upon successful return.*

4.5 Manage Queries

4.5.1 Description and Priority

The system shall allow block-wise managers to respond to customer queries and provide assistance.

***Purpose**: To improve communication and address customer concerns efficiently.*

***Priority**: Medium — Supports customer satisfaction but not core order flow.*

4.5.2 Stimulus/Response Sequences

Stimulus:

1. User selects "Contact Manager" from the menu.

2. User sends a query related to laundry services or order status.

3. Manager receives and replies to the query.

Response:

1. System stores the query and reply in the database.

2. System notifies the user once a reply is sent.

4.5.3 Functional Requirements

***FR1**: Provide a form for submitting queries to block-wise managers.*

***FR2**: Store queries and responses in the database.*

***FR3**: Notify managers upon query submission.*

***FR4**: Notify users when a reply is available*

5. Other Nonfunctional Requirements

5.1 Performance Requirements

System should handle up to 500 concurrent users.

Orders must be saved in under 3 seconds.

5.2 Safety Requirements

Automatic daily database backup.

Admin-only access for order modifications.

5.3 Security Requirements

Encrypted passwords.

Role-based access control.

5.4 Software Quality Attributes

Usability: Simple dashboard interface.

Reliability: 99% uptime.

Maintainability: Modular code structure.

5.5 Business Rules

Clothes must be returned exactly 2 days after order date.

No limit on clothes count.

Room number must be provided in every order.

6. Other Requirements

The system should support detergent inventory tracking.

Appendix A: Glossary

Gentle Wash: Mild detergent, low agitation wash cycle.

Daily Wash: Standard cleaning.

Hard Wash: Strong detergent, long wash cycle.

Appendix B: Analysis Models

ER diagram with entities: User, Admin, Order, Complaint, Detergent.

Appendix C: To Be Determined List

Finalize UI color scheme.

Confirm if SMS notification is needed.

Decide on detergent brand list.